Computational Genomics

EN.600.439 / 639

Fall 2015

Ben Langmead

langmea@cs.jhu.edu

www.langmead-lab.org

November 10, 2015

JOHNS HOPKINS
WHITING SCHOOL
of ENGINEERING

Department of Computer Science

Admin

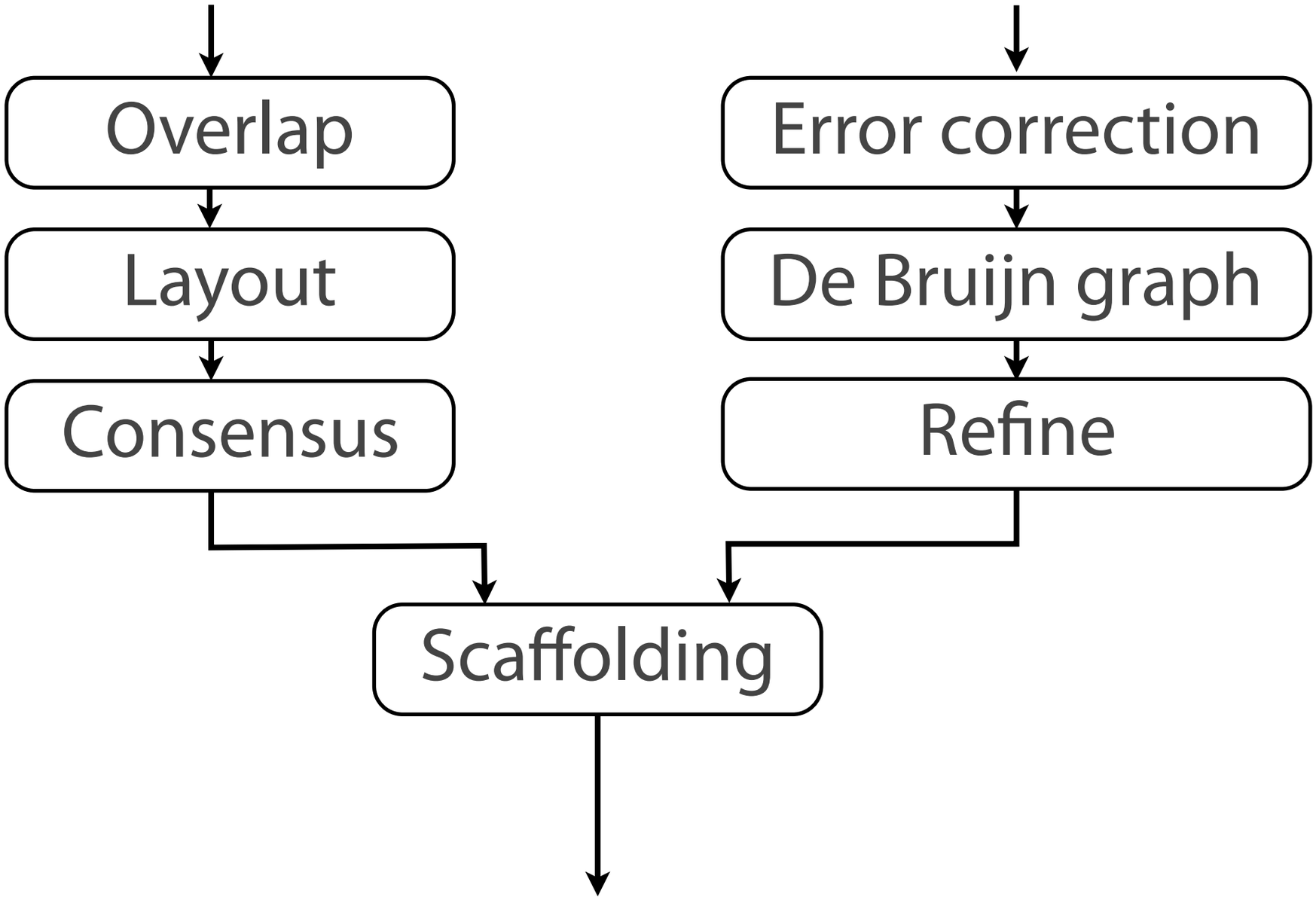
Homework 5 out, due 11/20 (not 11/19, which is first proj check-in)

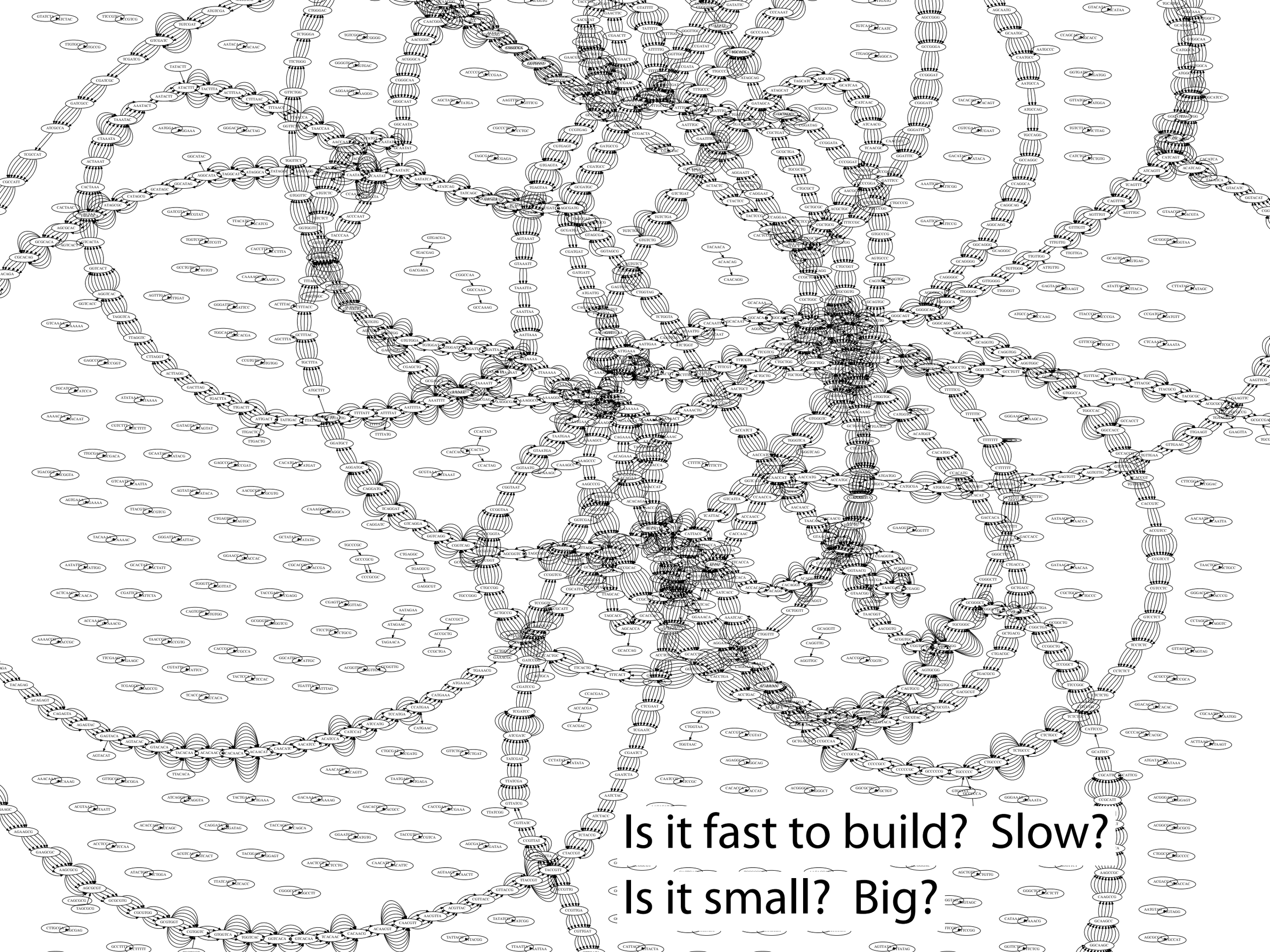
Homework 4 solutions posted, grades coming to you today

Assembly alternatives

Alternative 1: Overlap-Layout-Consensus (OLC) assembly

Alternative 2: De Bruijn graph (DBG) assembly



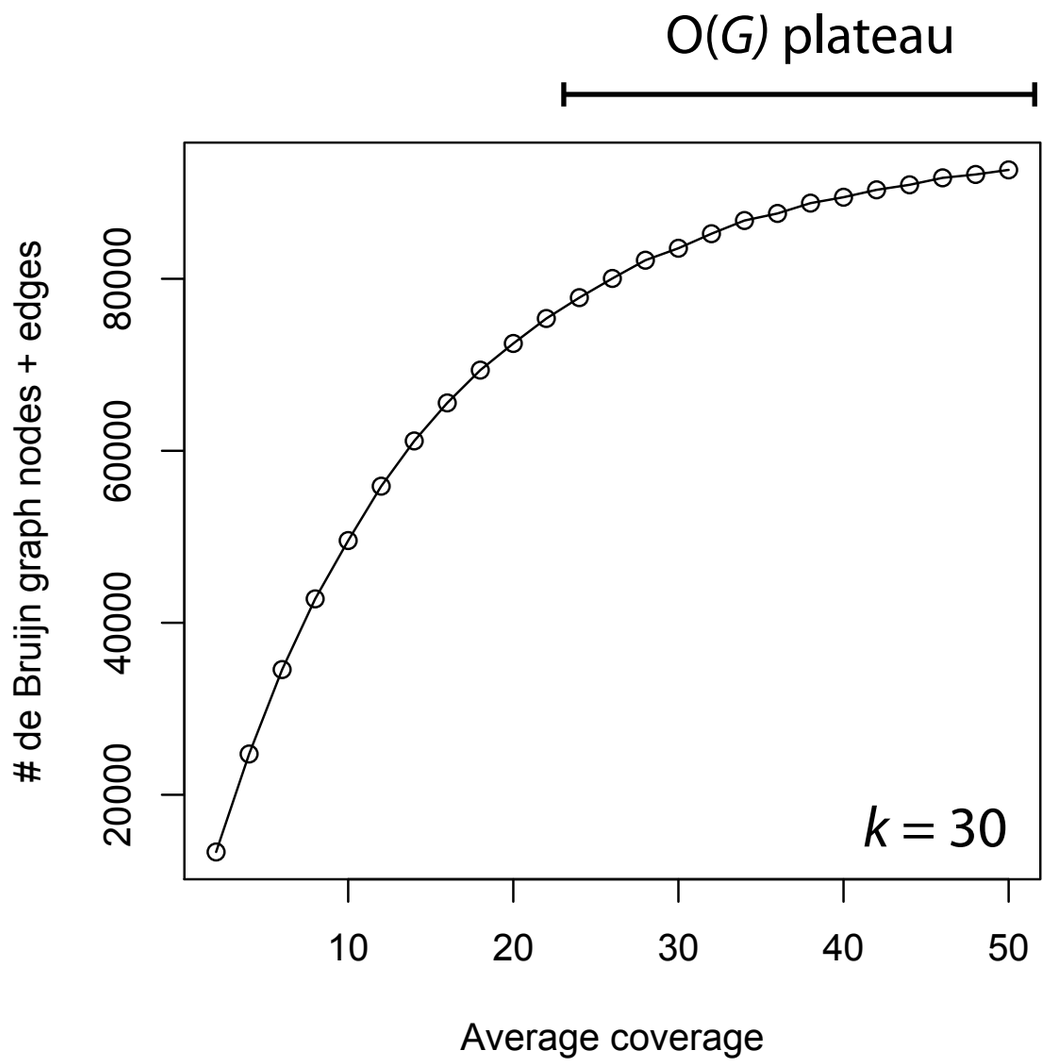


Is it fast to build? Slow?

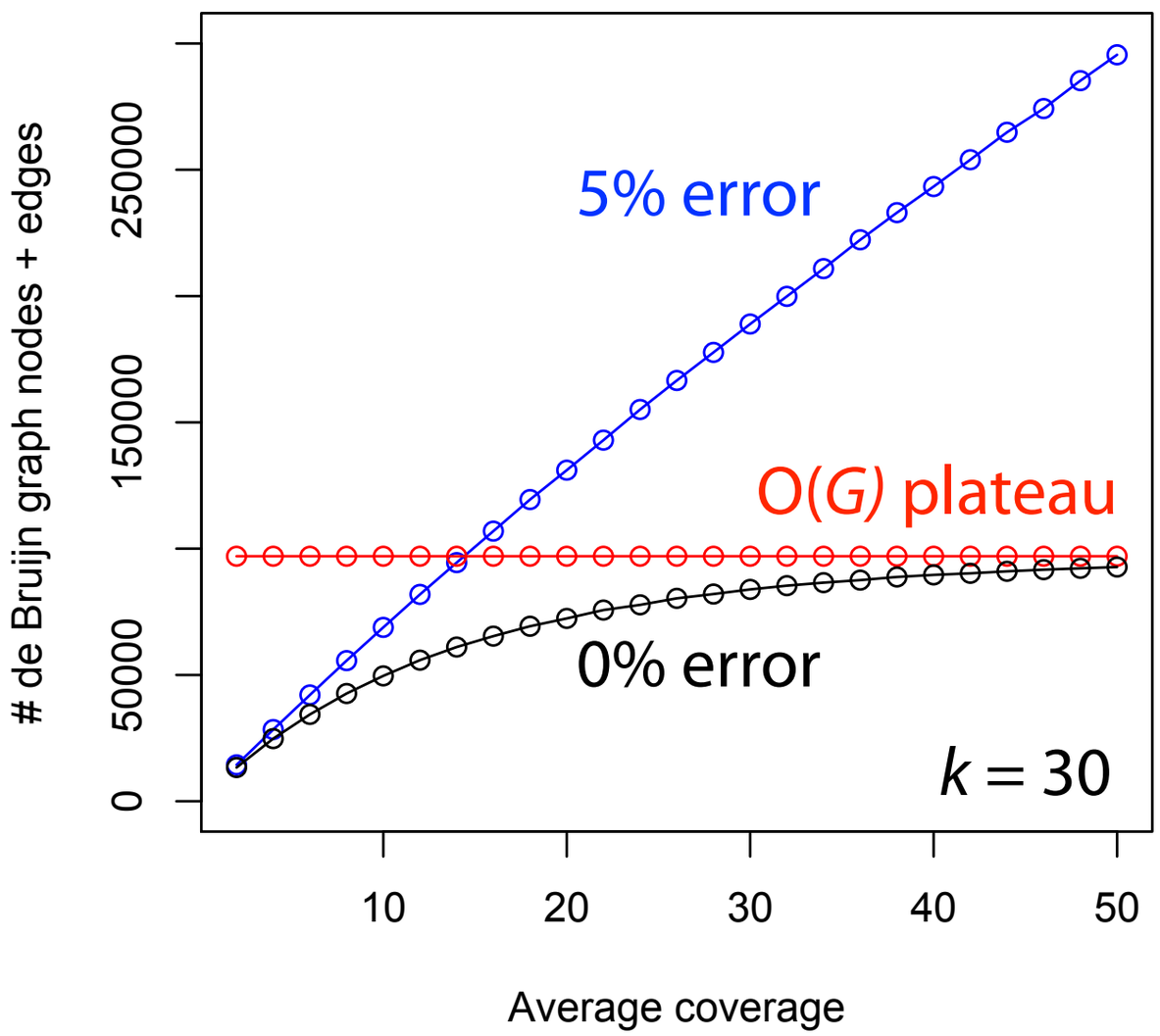
Is it small? Big?

De Bruijn graph

At high coverage, O(min(*N*, *G*)) bound is advantageous



Error correction



Lambda phage genome

Errors wipe out much of the benefit of the *G* bound

Error correction

For each read:

For each k-mer:

If *k*-mer count < *t*:

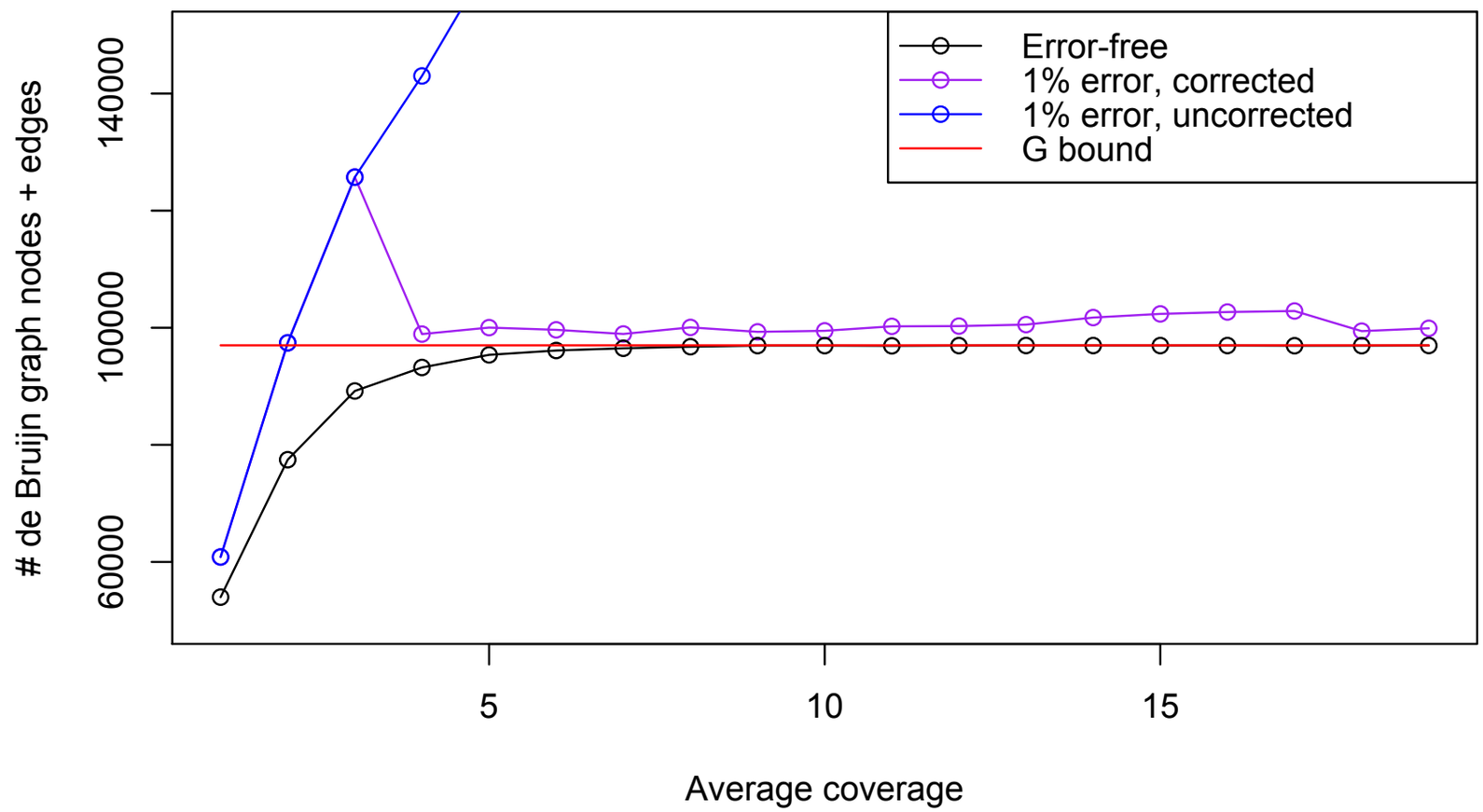
Examine *k*-mer's neighbors within certain Hamming/edit distance.
If neighbor has count ≥ *t*, replace old *k*-mer with neighbor.

http://bit.ly/CG_ErrorCorrect

Error correction: results

For uncorrected reads, De Bruijn graph size is off the chart

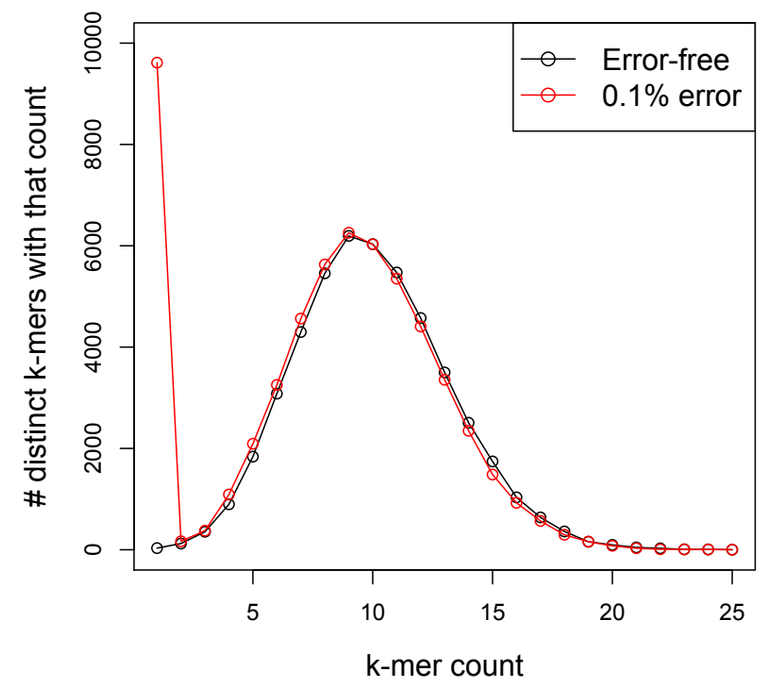
For corrected reads, De Bruijn graph size is near G bound



Error correction

For error correction to work well:

Average coverage and *k* must be set so that there's clear count separation between infrequent & frequent *k*-mers

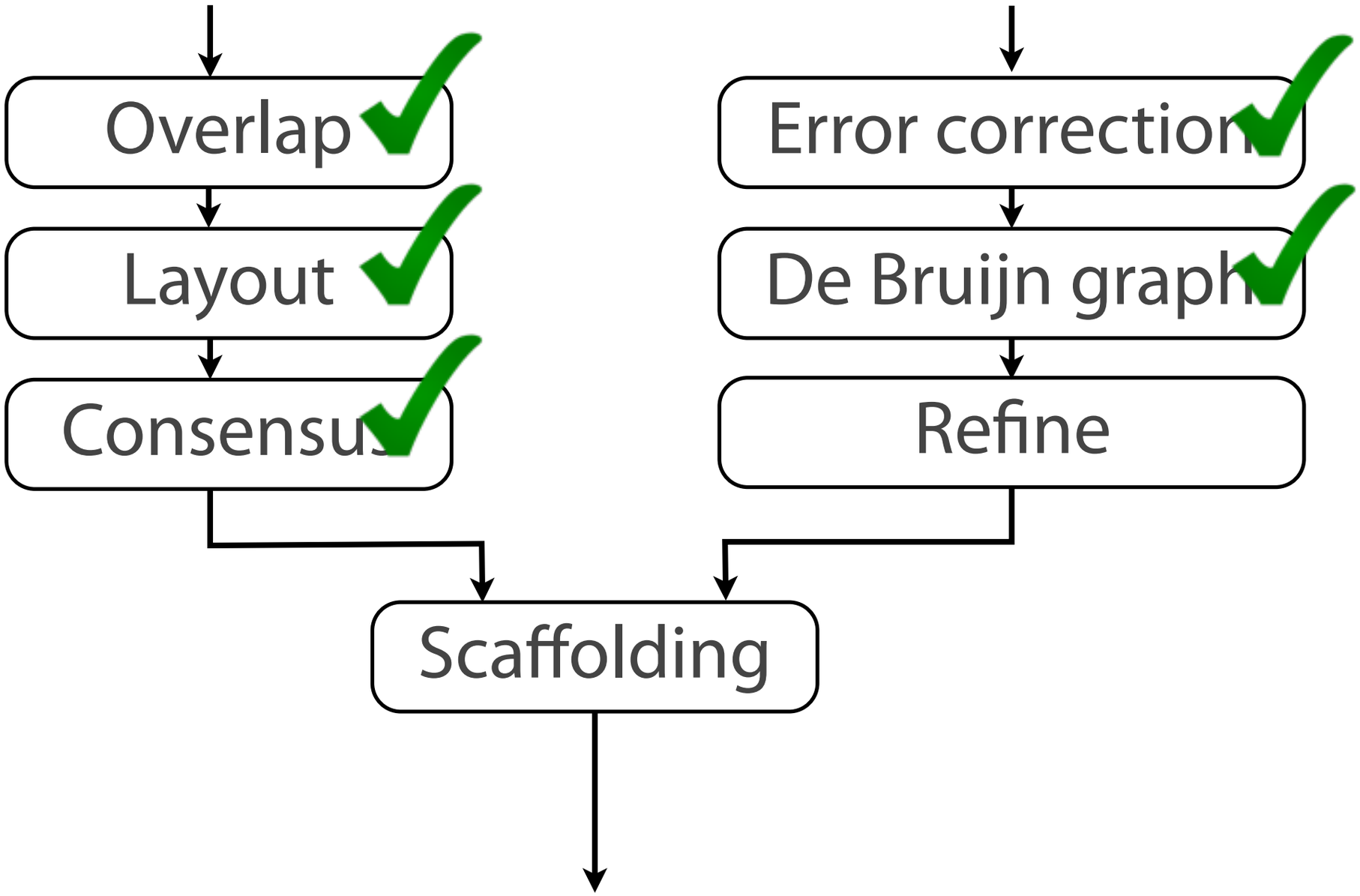


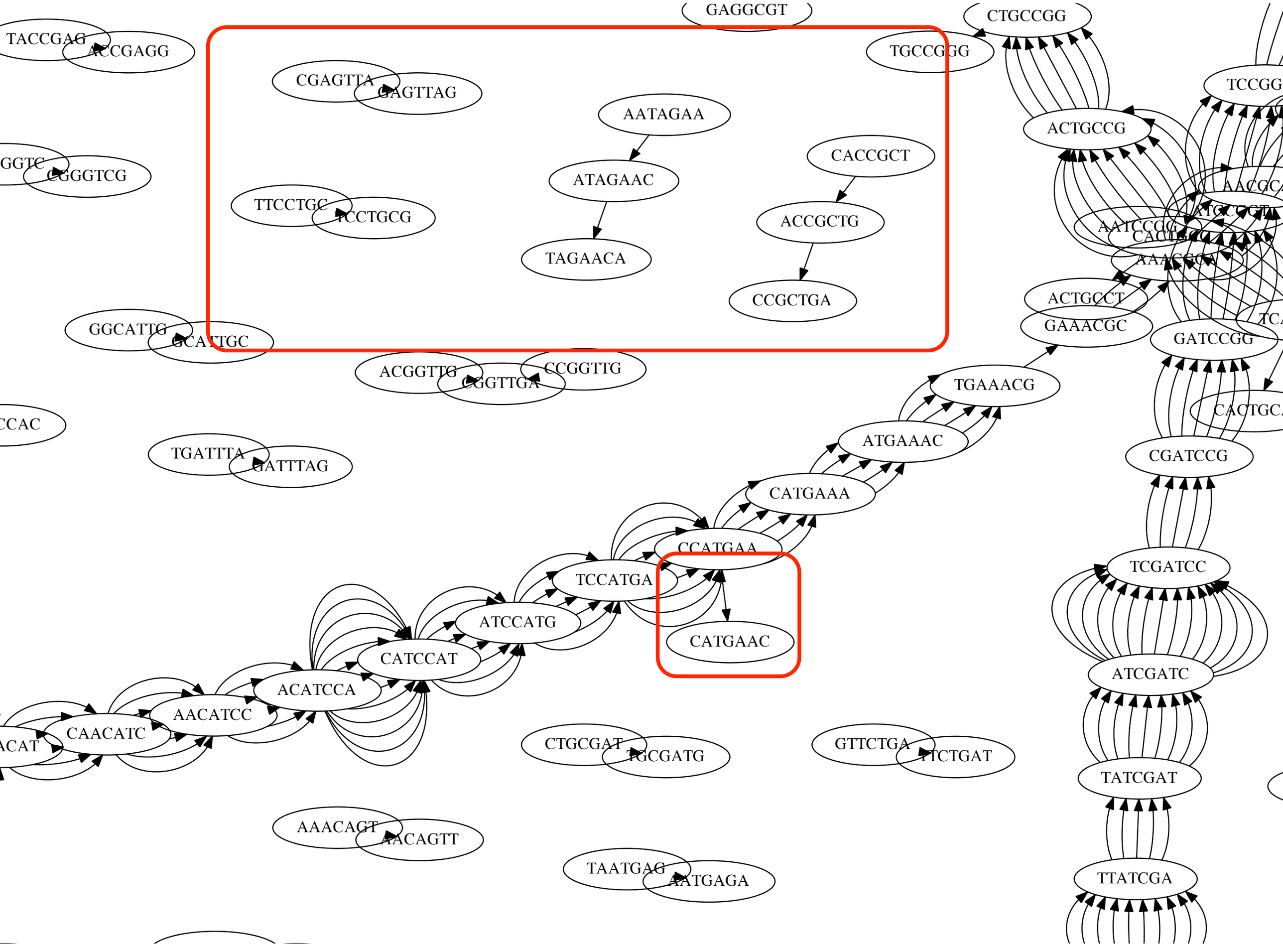
k-mer neighborhood explored must be broad enough to find frequent neighbors.

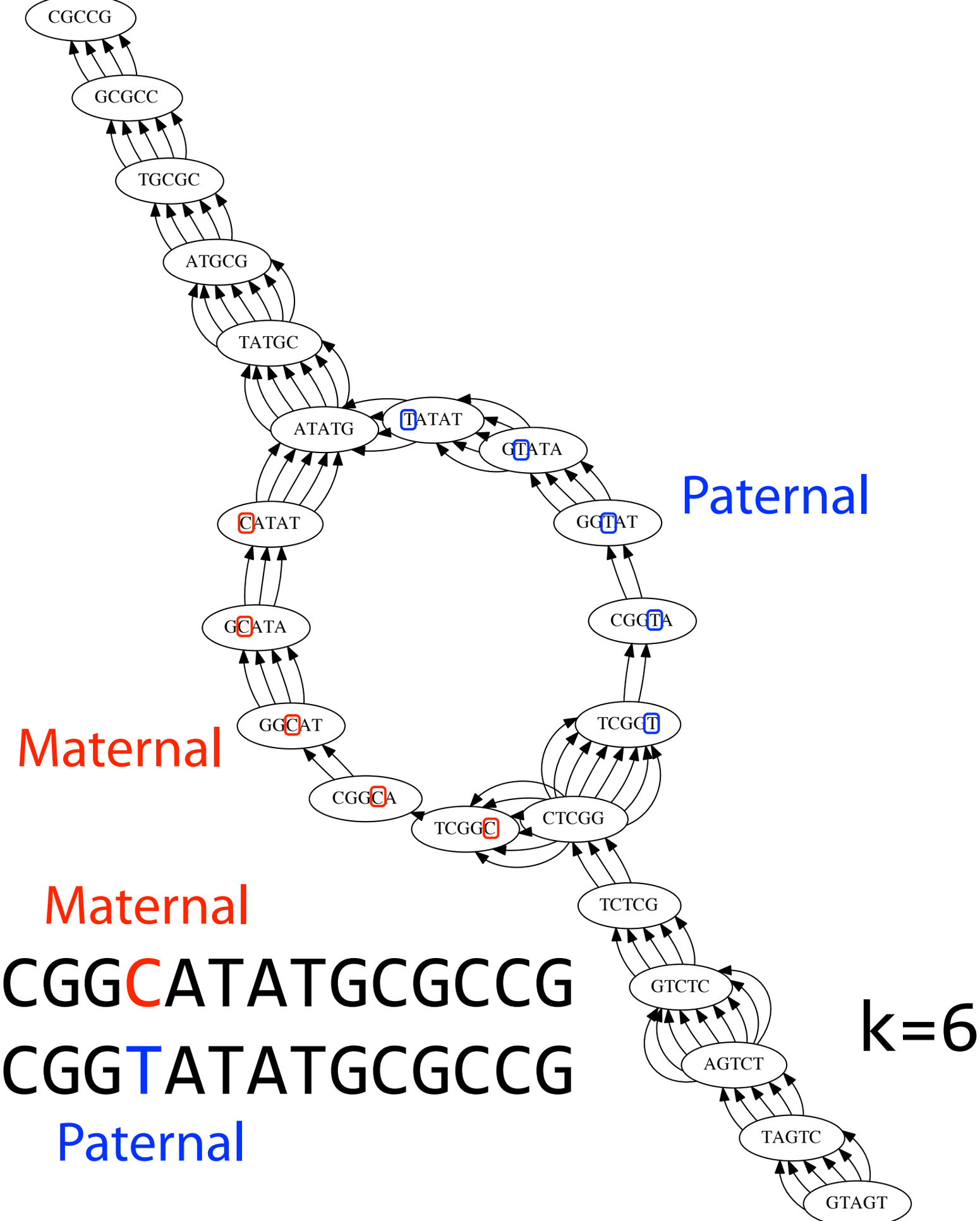
Data structure for storing *k*-mer counts should be smaller than De Bruijn graph (or what's the point?)

See: Bloom filters, CountMin, probabilistic counting, …

Assembly alternatives




GAGGCGT
CTGCCGG
TACCGAG
ACCGAGG
TGCCGGG
CGAGTTA
GAGTTAG
TCCGG
AATAGAA
ACTGCCG
CACCGCT
GGTC
CGGGTCG
ATAGAAC
AACGC
TTCCTGC
TCCTGCG
ACCGCTG
AATCCGG
TAGAACA
ACTGCCT
CCGCTGA
GAAACGC
GGCATTG
GCATTGC
GATCCGG
ACGGTTG
CCGGTTG
CGGTTGA
TGAAACG
CCAC
ATGAAAC
TGATTTA
GATTTAG
CGATCCG
CATGAAA
CCATGAA
TCCATGA
TCGATCC
ATCCATG
CATGAAC
CATCCAT
ATCGATC
ACATCCA
AACATCC
CAACATC
CTGCGAT
TGCGATG
GTTCTGA
TTCTGAT
TATCGAT
AAACAGT
AACAGTT
TAATGAG
AATGAGA
TTATCGA



Maternal

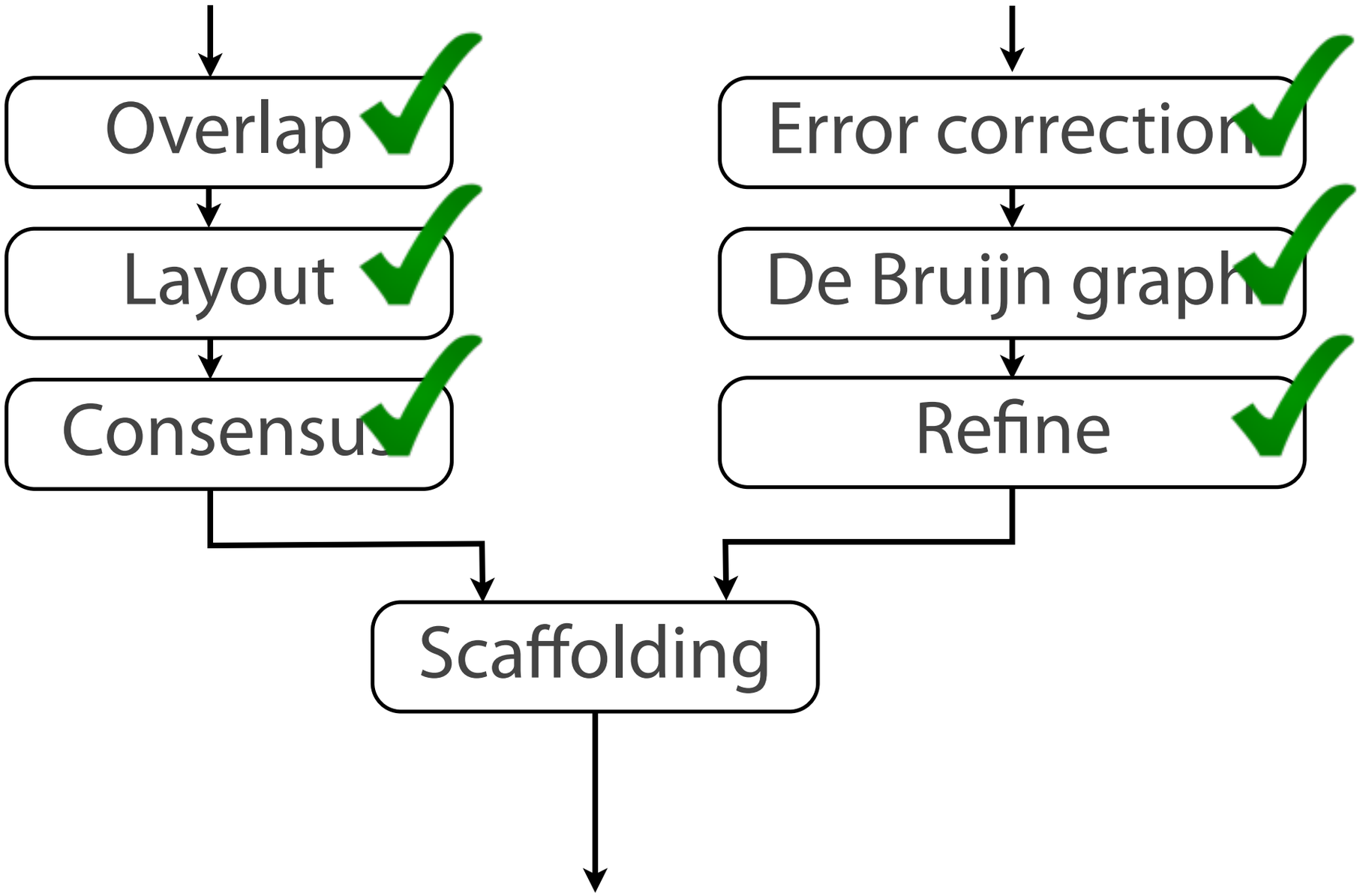
GTAGTCTCGGCATATGCGCCG

GTAGTCTCGGTATATGCGCCG

Paternal

k=6

Assembly alternatives



Scaffolding

Once we've built contigs, scaffolding *orders* and *orients* contigs with respect to each other

We can use data from various sources, but especially *paired ends*

Scaffolding: paired-end sequencing

Produces a *pair* of reads taken from either end of a longer *fragment*

Scaffolding

Scaffolding output: collection of *scaffolds*, where a scaffold is a collection of contigs related to each other with high confidence using pairs

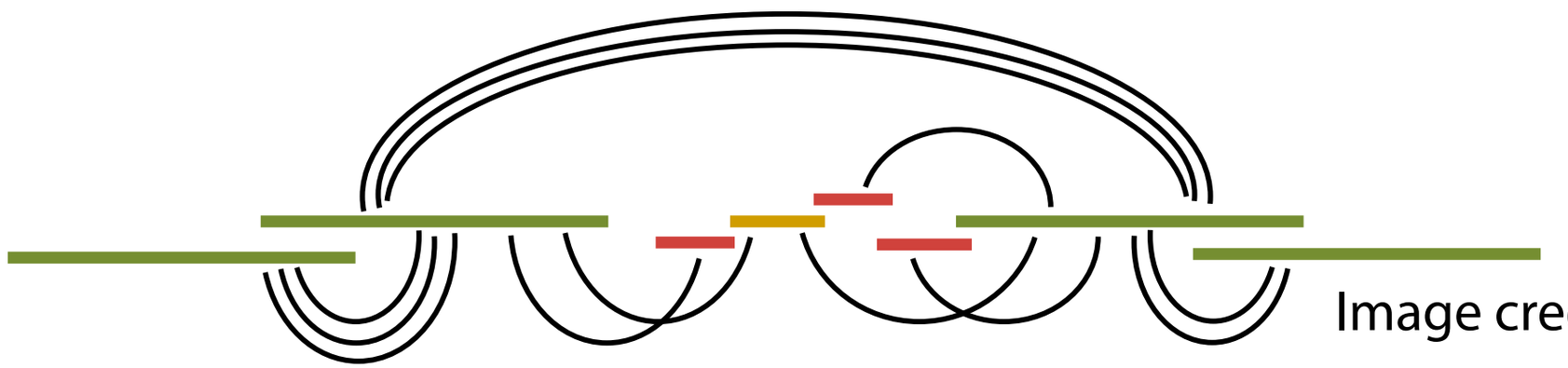
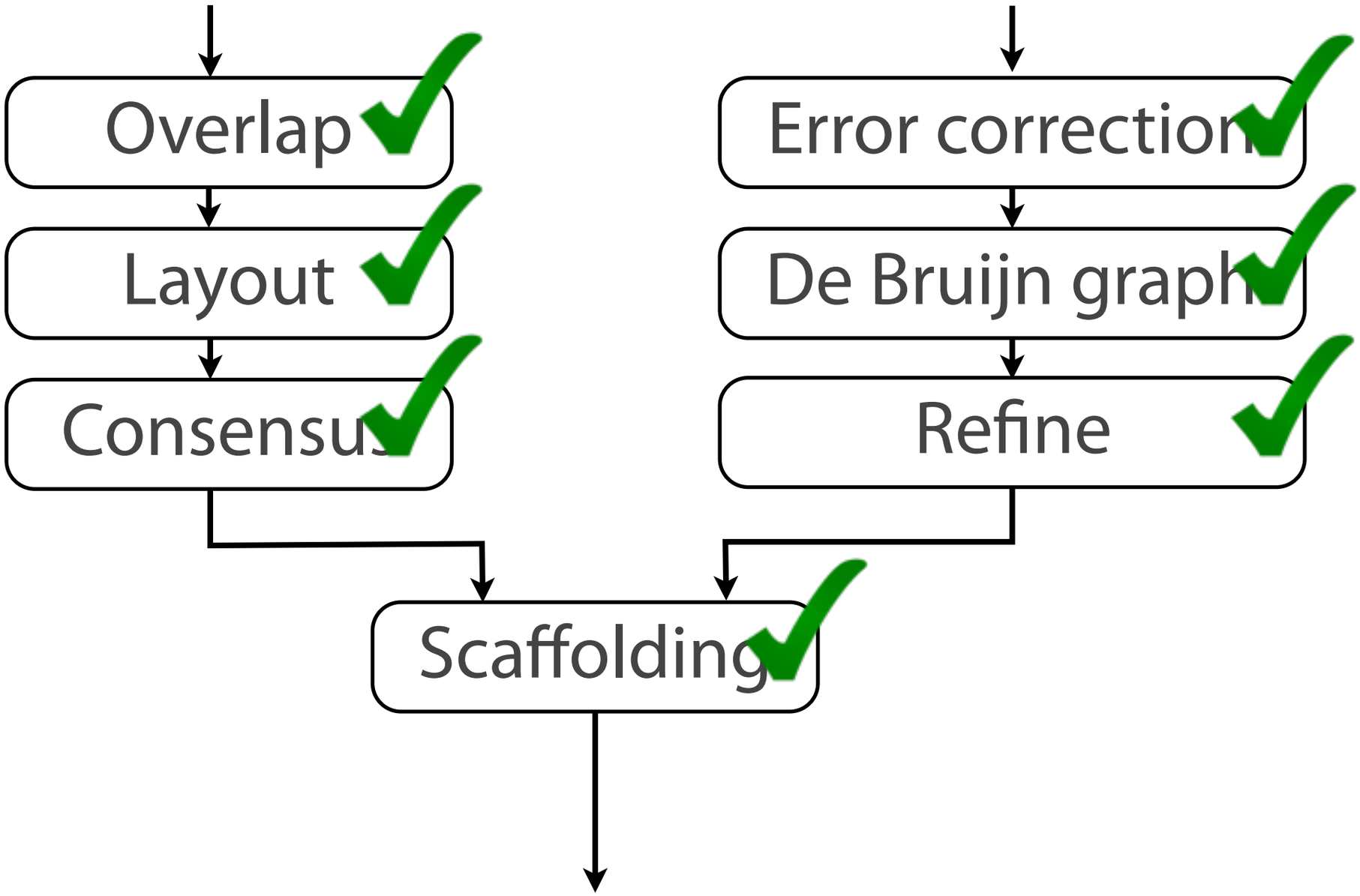


Image credit: Mike Schatz

Assembly alternatives



Real-world assemblers

Overlap-Layout-Consensus

Celera Assembler: ~ 1,100 citations

Used for commercial human genome project, fruitfly, cow, strawberry, many others; still used today

wgs-assembler.sourceforge.net

Gene Myers, et al. "A whole-genome assembly of Drosophila." *Science* 287.5461 (2000): 2196-2204.

ARACHNE: ~ 550 citations

Used for Mouse, Chimpanzee, Dog, Horse genomes

www.broadinstitute.org/crd/wiki/index.php/Arachne_Main_Page

Serafim Batzoglou, et al. "ARACHNE: a whole-genome shotgun assembler." *Genome research* 12.1 (2002): 177-189.

Real-world assemblers

De Bruijn graph

Velvet: ~ 4,000 citations

ABySS: ~ 1,400 citations

SOAPdenovo: ~ 1,300 citations

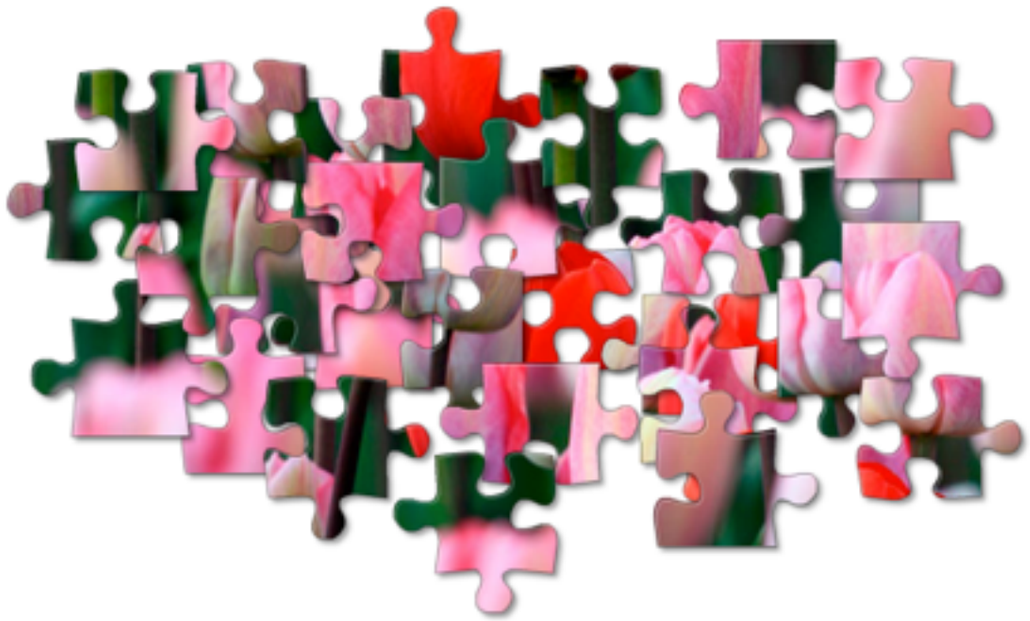
ALLPATHS: ~ 500 citations

EULER-SR: ~ 400 citations

Efforts to describe and evaluate these tools:

http://assemblathon.org

http://gage.cbcb.umd.edu/index.html



+

chr11:5246500-5248500 (reverse strand):

ATATCTTAGAGGGAGGGCTGAGGGTTTGAAGTCCAACTCCTAAGCCAGTGCCAGAAGAGCCAAGGACAGGTACGGCTGTC
ATCACTTAGACCTCACCCTGTGGAGCCACACCCTAGGGTTGGCCAATCTACTCCCAGGAGCAGGGAGGGCAGGAGCCAGG
GCTGGGCATAAAAGTCAGGGCAGAGCCATCTATTGCTTACATTTGCTTCTGACACAACTGTGTTCACTAGCAACCTCAAA
CAGACACCATGGTGCATCTGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAGTT
GGTGGTGAGGCCCTGGGCAGGTTGGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCATGTGGAGA
CAGAGAAGACTCTTGGGTTTCTGATAGGCACTGACTCTCTCTGCCTATTGGTCTATTTTCCCACCCTTAGGCTGCTGGTG
GTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATGCTGTTATGGGCAACCCTAAGGT
GAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTCACCTGGACAACCTCAAGGGCACCTTTGCCA
CACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGATCCTGAGAACTTCAGGGTGAGTCTATGGGACGCTTGATGTTTT
CTTTCCCCTTCTTTTCTATGGTTAAGTTCATGTCATAGGAAGGGGATAAGTAACAGGGTACAGTTTAGAATGGGAAACAG
ACGAATGATTGCATCAGTGTGGAAGTCTCAGGATCGTTTTAGTTTCTTTTATTTGCTGTTCATAACAATTGTTTTCTTTT
GTTTAATTCTTGCTTTCTTTTTTTTTCTTCTCCGCAATTTTTACTATTATACTTAATGCCTTAACATTGTGTATAACAAA
AGGAAATATCTCTGAGATACATTAAGTAACTTAAAAAAAAACTTTACACAGTCTGCCTAGTACATTACTATTTGGAATAT
ATGTGTGCTTATTTGCATATTCATAATCTCCCTACTTTATTTTCTTTTATTTTTAATTGATACATAATCATTATACATAT
TTATGGGTTAAAGTGTAATGTTTTAATATGTGTACACATATTGACCAAATCAGGGTAATTTTGCATTTGTAATTTTAAAA
AATGCTTTCTTCTTTTAATATACTTTTTTGTTTATCTTATTTCTAATACTTTCCCTAATCTCTTTCTTTCAGGGCAATAA
TGATACAATGTATCATGCCTCTTTGCACCATTCTAAAGAATAACAGTGATAATTTCTGGGTTAAGGCAATAGCAATATCT
CTGCATATAAATATTTCTGCATATAAATTGTAACTGATGTAAGAGGTTTCATATTGCTAATAGCAGCTACAATCCAGCTA
CCATTCTGCTTTTATTTTATGGTTGGGATAAGGCTGGATTATTCTGAGTCCAAGCTAGGCCCTTTTGCTAATCATGTTCA
TACCTCTTATCTTCCTCCCACAGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTCACC
CCACCAGTGCAGGCTGCCTATCAGAAAGTGGTGGCTGGTGTGGCTAATGCCCTGGCCCACAAGTATCACTAAGCTCGCTT
TCTTGCTGTCCAATTTCTATTAAAGGTTCCTTTGTTCCCTAAGTCCAACTACTAAACTGGGGGATATTATGAAGGGCCTT
GAGCATCTGGATTCTGCCTAATAAAAAACATTTATTTTCATTGCAATGATGTATTTAAATTATTTCTGAATATTTTACTA
AAAAGGGAATGTGGGAGGTCAGTGCATTTAAAACATAAAGAAATGAAGAGCTAGTTCAAACCTTGGGAAAATACACTATA
TCTTAAACTCCATGAAAGAAGGTGAGGCTGCAAACAGCTAATGCACATTGGCAACAGCCCCTGATGCATATGCCTTATTC

chr11:5246500-5248500 (reverse strand):

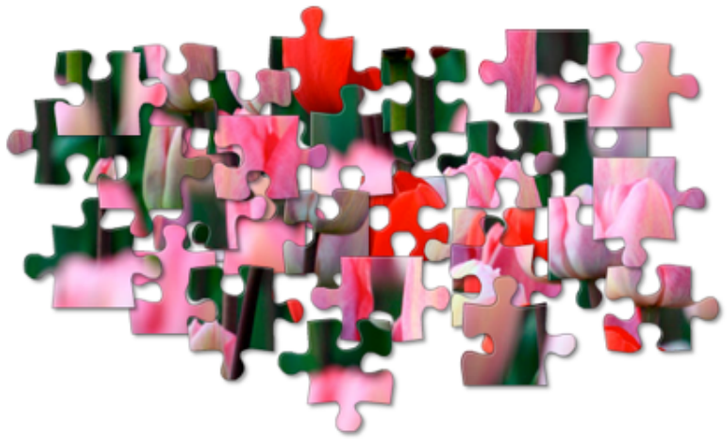
Sickle cell anemia

Homo sapiens hemoglobin, beta (HBB)

```
ATATCTTAGAGGGAGGGCTGAGGGTTTGAA                                  CAGAAGAGCCAAGGACAGGTACGGCTGTC
ATCACTTAGACCTCACCCTGTGGAGCCACA                                  TCCCAGGAGCAGGGAGGGCAGGAGCCAGG
GCTGGGCATAAAAGTCAGGGCAGAGCCATCTATTGCTTACATTTGCTTCTGACACAACTGTGTTCACTAGCAACCTCAAA
CAGACACCATGGTGCATCTGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAGTT
GGTGGTGAGGCCCTGGGCAGGTTGGTATCAAGGTTACAAGACAGGTTTAAGGAGACCAATAGAAACTGGGCATGTGGAGA
CAGAGAAGACTCTTGGGTTTCTGATAGGCACTGACTCTCTCTGCCTATTGGTCTATTTTCCCACCCTTAGGCTGCTGGTG
GTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATGCTGTTATGGGCAACCCTAAGGT
GAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTCACCTGGACAACCTCAAGGGCACCTTTGCCA
CACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGATCCTGAGAACTTCAGGGTGAGTCTATGGGACGCTTGATGTTTT
CTTTCCCCTTCTTTTCTATGGTTAAGTTCATGTCATAGGAAGGGGATAAGTAACAGGGTACAGTTTAGAATGGGAAACAG
ACGAATGATTGCATCAGTGTGGAAGTCTCAGGATCGTTTTAGTTTCTTTTATTTGCTGTTCATAACAATTGTTTTCTTTT
GTTTAATTCTTGCTTTCTTTTTTTTTCTTCTCCGCAATTTTTACTATTATACTTAATGCCTTAACATTGTGTATAACAAA
AGGAAATATCTCTGAGATACATTAAGTAACTTAAAAAAAAACTTTACACAGTCTGCCTAGTACATTACTATTTGGAATAT
ATGTGTGCTTATTTGCAT                                              TACATAATCATTATACATAT
TTATGGGTTAAAGTGTAA                                              TTGCATTTGTAATTTTAAAA
AATGCTTTCTTCTTTTAATATACTTTTTTGTTTATCTTATTTCTAATACTTTCCCTAATCTCTTTCTTTCAGGGCAATAA
TGATACAATGTATCATGCCTCTTTGCACCATTCTAAAGAATAACAGTGATAATTTCTGGGTTAAGGCAATAGCAATATCT
CTGCATATAAATATTTCTGCATATAAATTGTAACTGATGTAAGAGGTTTCATATTGCTAATAGCAGCTACAATCCAGCTA
CCATTCTGCTTTTATTTTATGGTTGGGATAAGGCTGGATTATTCTGAGTCCAAGCTAGGCCCTTTTGCTAATCATGTTCA
TACCTCTTATCTTCCTCCCACAGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTCACC
CCACCAGTGCAGGCTGCCTATCAGAAAGTGGTGGCTGGTGTGGCTAATGCCCTGGCCCACAAGTATCACTAAGCTCGCTT
TCTTGCTGTCCAATTTCTATTAAAGGTTCCTTTGTTCCCTAAGTCCAACTACTAAACTGGGGGATATTATGAAGGGCCTT
GAGCATCTGGATTCTGCCTAATAAAAAACATTTATTTTCATTGCAATGATGTATTTAAATTATTTCTGAATATTTTACTA
AAAAGGGAATGTGGGAGGTCAGTGCATTTAAAACATAAAGAAATGAAGAGCTAGTTCAAACCTTGGGAAAATACACTATA
TCTTAAACTCCATGAAAGAAGGTGAGGCTGCAAACAGCTAATGCACATTGGCAACAGCCCCTGATGCATATGCCTTATTC
```

Picking up signals

So far, we've focused on how to stitch fragmentary evidence into bigger pictures, i.e. genomes



Now we have more questions!

Where are the genes?

Where/what is the *functional* DNA?

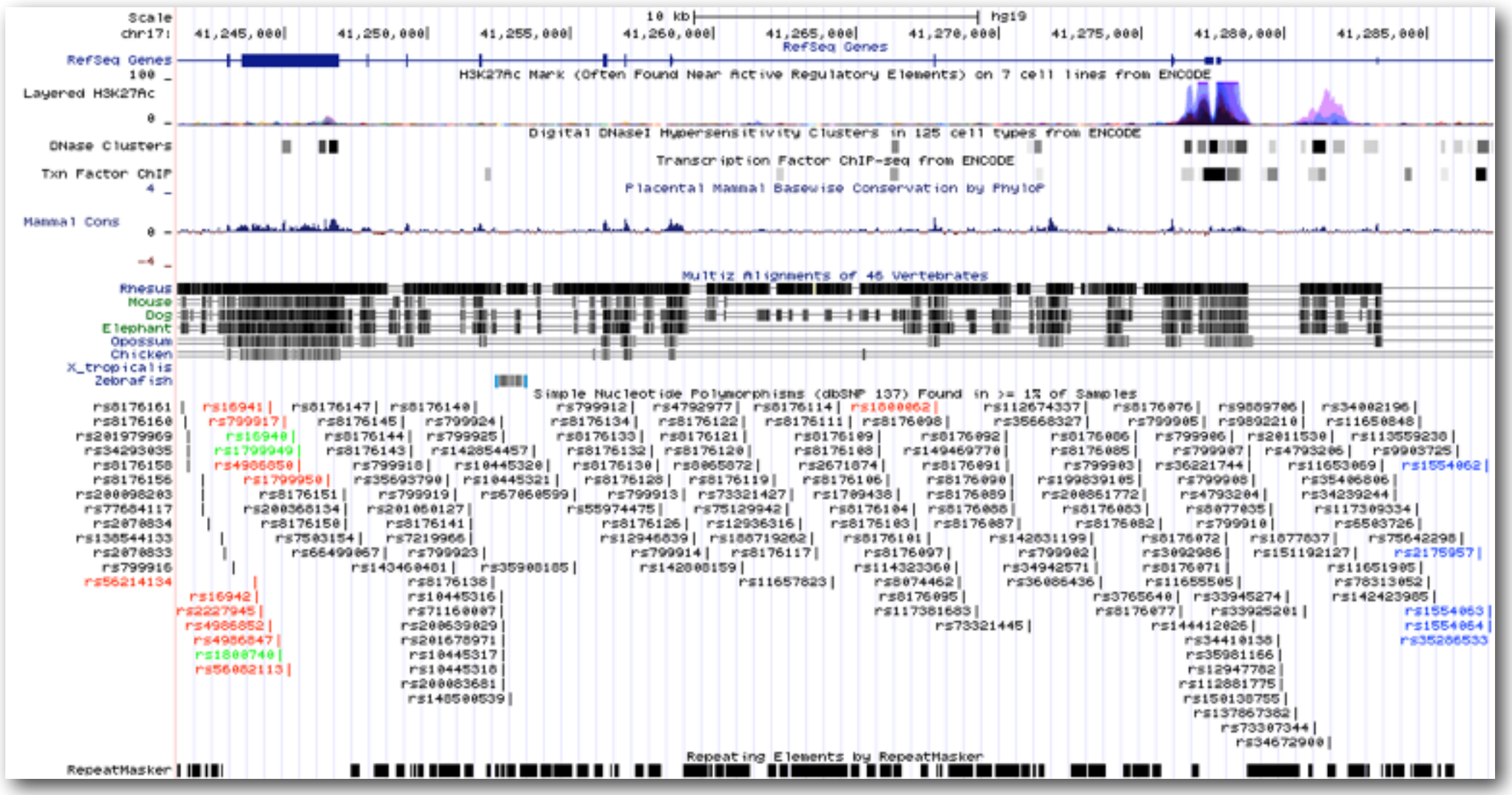
What's different about the DNA in different tissues?

In what abundance do we find various molecules?

What differences exist between individuals?

Picking up signals

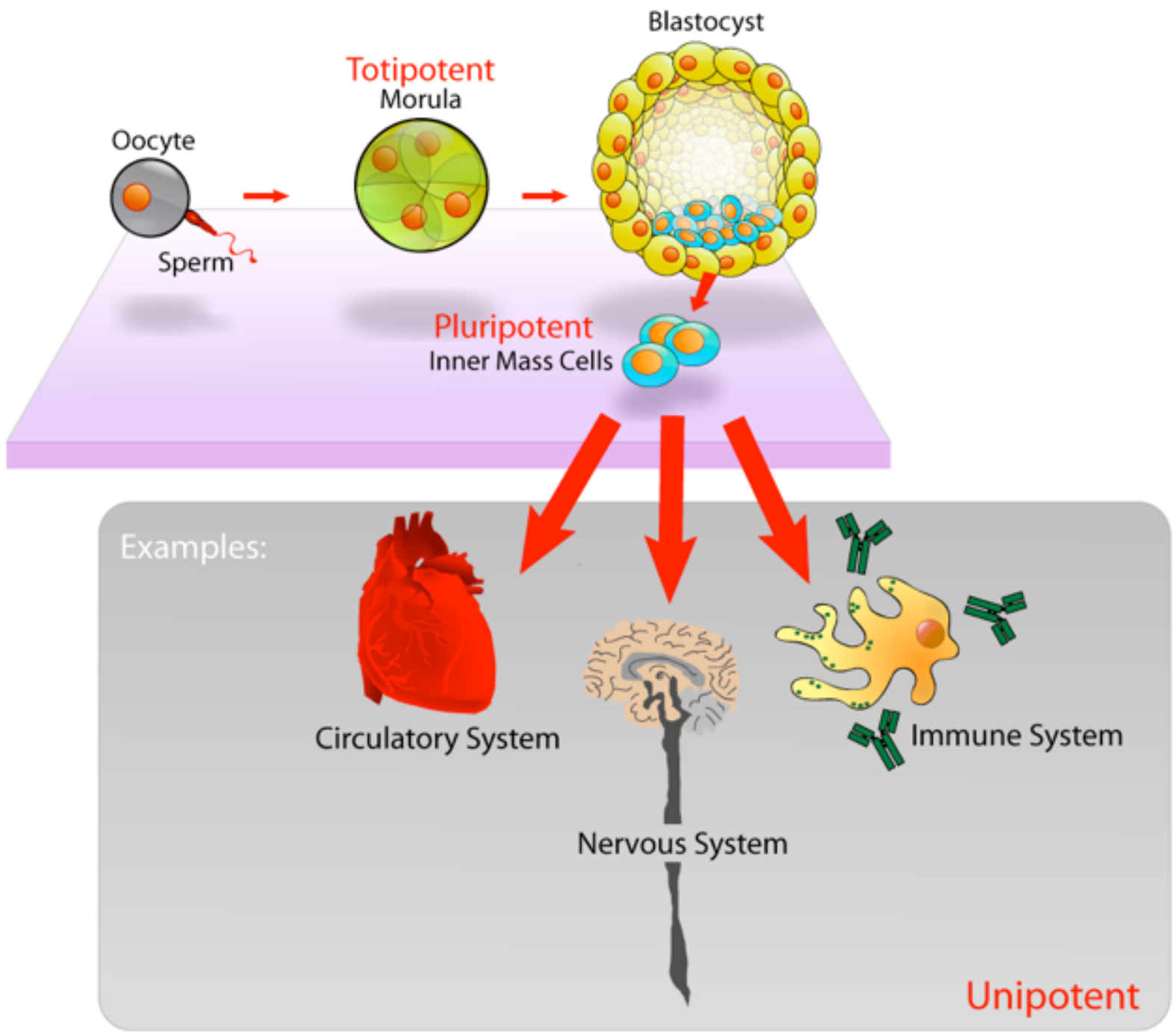
Through many experiments, we know much more about the genome than just its DNA sequence:



40 K nt region of chromosome 17

http://genome.ucsc.edu/cgi-bin/hgTracks

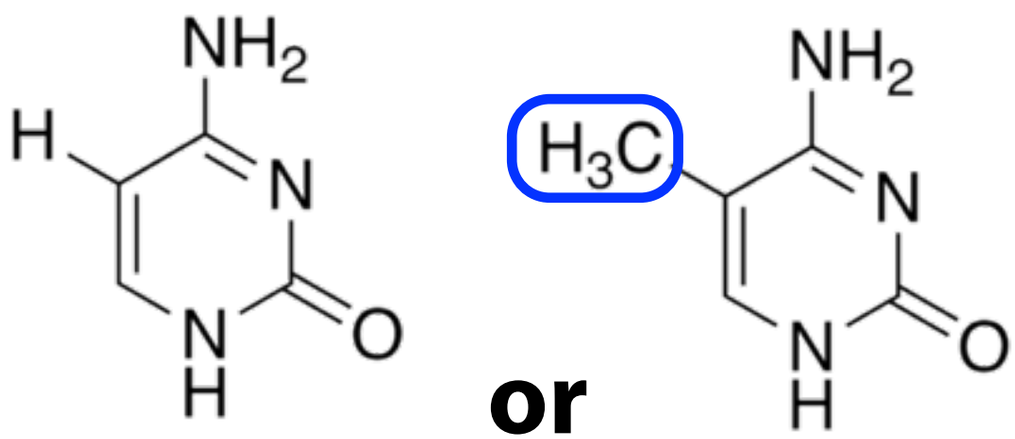
Epigenetics



http://en.wikipedia.org/wiki/File:Stem_cells_diagram.png

CpG Islands

Dinucleotide "CG" (AKA "CpG") is special because C can have a *methyl group* attached

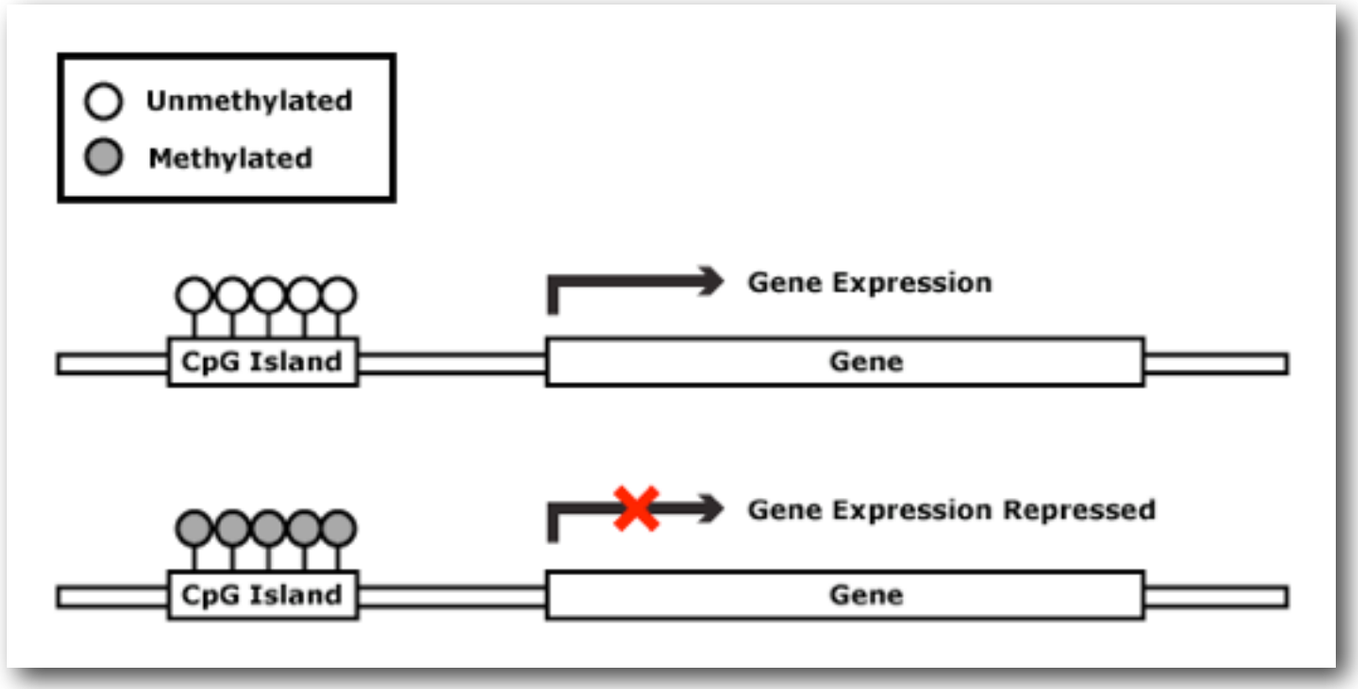


Unmethylated Methylated

CpG Islands

CpG island: part of the genome where CG occurs particularly frequently

Usually regulate expression of nearby genes



Cells from different tissues have different patterns of CpG methylation and different gene expression profiles

Key *epigenetic* phenomenon

http://missinglink.ucsf.edu/lm/genes_and_genomes/methylation.html

CpG Islands

Let's design a scheme for scoring a k-mer according to how confident we are it belongs to a CpG island

Ideally, scores should be based on *probabilities*

Probability mini-review

Sample space (Ω) is set of all possible outcomes

E.g. Ω = { all possible rolls of 2 dice }

An *event* (A, B, C, ...) is a subset of Ω

A = { rolls where first die is odd }, B = { rolls where second die is even }

We're often concerned with assigning a probability to an event

P(A): fraction of all possible outcomes that are in A

$P(A) = |A| / |\Omega| = 18 / 36 = 0.5$

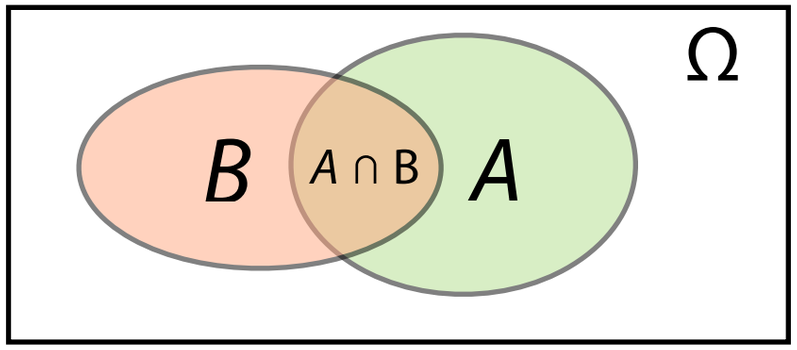
Probability mini-review

P(A, B): fraction of all possible outcomes in both A and B

$P(A, B) = |A \cap B| / |\Omega| = 9 / 36 = 0.25$

Also written: $P(A \cap B)$ or $P(AB)$

Joint probability of A and B



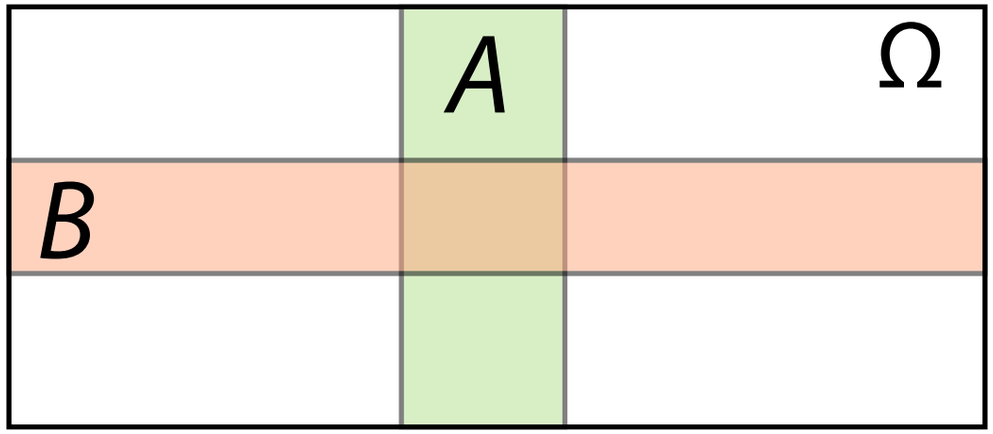
P(A | B): fraction of outcomes in B that are also in A

$P(A \mid B) = |A \cap B| / |B| = 9 / 18 = 0.5$

$P(A \mid B) = P(A, B) / P(B)$

$P(A, B) = P(A \mid B) \cdot P(B)$

Probability mini-review



Events A and B are independent if $\mathrm{P}(A \mid B) = \mathrm{P}(A)$

So $\mathrm{P}(A, B) = \mathrm{P}(B)\,\mathrm{P}(A \mid B) = \mathrm{P}(A)\,\mathrm{P}(B)$

Probability mini-review

More probability review, courtesy of Prof. Joe Blitzstein and others:

http://projects.iq.harvard.edu/stat110/youtube

http://j.mp/CG_prob_cheatsheet

Sequence models

Sequence model is a *probabilistic model* that associates probabilities with sequences

What *k*-mers do I see inside versus outside of a CpG island?

What's the probability of next character being A if previous characters were GATTAC?

Given a genome, where are the genes?

Right: model for eukaryotic gene finding

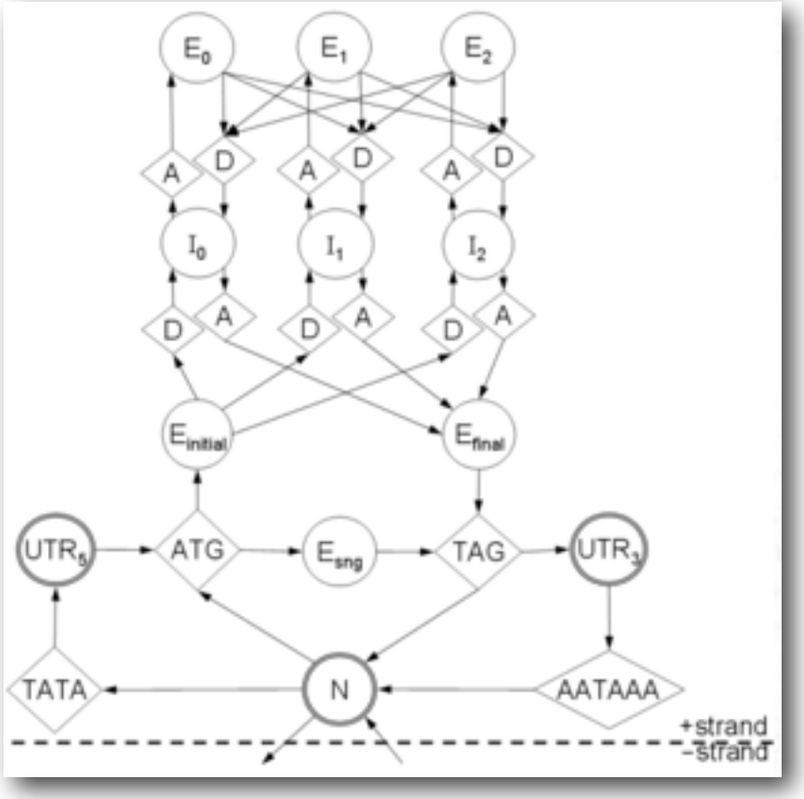


Image by Bill Majoros:
http://www.genezilla.org/design.html

Sequence models

Sequence models *learn from examples*

Say we have sampled 100K 5-mers from inside CpG islands and 100K 5-mers from outside

Can we guess whether CGCGC came from a CpG island?

# CGCGC inside	315
# CGCGC outside	12

p(inside) = 315/(315 + 12)= 0.963

http://bit.ly/CG_MarkovChain

Sequence models

P(x) = probability of sequence x

$$P(x) = \underbrace{P(x_k, x_{k-1}, \ldots x_1)}_{\text{Joint probability of each base}}$$

Estimating P(x): # occurrences of x *inside* divided by # occurrences overall

For larger k, might see few or no occurrences of x. Joint probabilities for rare events are hard to estimate well!

Sequence models

$$P(x) = P(x_k, x_{k-1}, \ldots x_1)$$

Re-write using conditional probability:

$$= P(x_k \mid x_{k-1}, \ldots x_1)\, P(x_{k-1}, \ldots x_1)$$

$$= P(x_k \mid x_{k-1}, \ldots x_1)\, P(x_{k-1} \mid x_{k-2}, \ldots x_1)\, P(x_{k-2}, \ldots x_1)$$

(etc)

Simplifying assumption: probability of a particular item at position k depends only on item at previous position: x_{k-1}

Technically: x_k is *conditionally independent* of $x_1 \ldots x_{k-2}$ given x_{k-1}

Informally: "the future is independent of the past given the present"

Sequence models

Simplifying assumption: probability of a particular item at position k depends only on item at previous postion: x_{k-1}

$$
\begin{aligned}
P(x) &= P(x_k, x_{k-1}, \ldots x_1) \\
&= P(x_k \mid x_{k-1}, \ldots x_1)\, P(x_{k-1}, \ldots x_1) \\
&= P(x_k \mid x_{k-1}, \ldots x_1)\, P(x_{k-1} \mid x_{k-2}, \ldots x_1)\, P(x_{k-2}, \ldots x_1)
\end{aligned}
$$

(etc) drop drop (etc)

$$\approx P(x_k \mid x_{k-1})\, P(x_{k-1} \mid x_{k-2}) \ldots P(x_2 \mid x_1)\, P(x_1)$$

Markov property / assumption

Big assumption, but often reasonable and opens the door to tractable, powerful algorithms

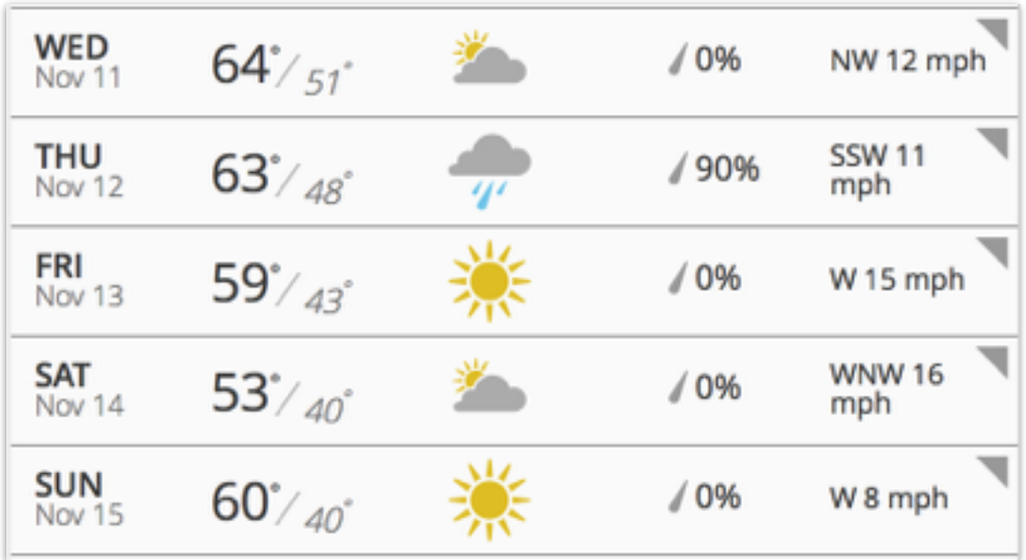
Markov assumption

"To predict next state of the Parcheesi game, just tell me the current state. I don't care about any other previous states."

Reasonable assumption; basically true

"To predict today's weather, just tell me yesterday's weather. I don't care about any other previous days' weather."

Actually, it clearly helps more to know about more previous days. But still might be a useful assumption.



WED Nov 11	64°/51°		0%	NW 12 mph
THU Nov 12	63°/48°		90%	SSW 11 mph
FRI Nov 13	59°/43°		0%	W 15 mph
SAT Nov 14	53°/40°		0%	WNW 16 mph
SUN Nov 15	60°/40°		0%	W 8 mph

Markov chain

Assigning a probability to a sequence using Markov property:

$$P(x) \underset{\text{Markov property}}{\approx} P(x_k \mid x_{k-1})\, P(x_{k-1} \mid x_{k-2}) \ldots P(x_2 \mid x_1)\, P(x_1)$$

Say x is a nucleotide k-mer

$P(x_i \mid x_{i-1})$ probability of seeing nucleotide x_i in i^{th} position given that previous nucleotide is x_{i-1}

Shorthand: $P(G \mid C)$ = probability of G given previous is C

Markov chain

Say we are given sequences of several CpG islands. How do we estimate, say, $P(G \mid C)$?

$$P(G \mid C) = \text{\# times CG occurs / \# times C occurs}$$

Markov chain

Given CpG island sequences from human chromosome 1, count nucleotide and dinucleotide occurrences and estimate all 16 possible $P(x_i \mid x_{i-1})$:

$P(A \mid A)$ = # times AA occurs / # times A occurs

$P(C \mid A)$ = # times AC occurs / # times A occurs

$P(G \mid A)$ = # times AG occurs / # times A occurs

$P(T \mid A)$ = # times AT occurs / # times A occurs

$P(A \mid C)$ = # times CA occurs / # times C occurs

(etc)